IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : R26-SE-022

1. Topic (12 words max)

An AI-Enabled Intelligent Study Assistant to Enhance Student Learning Outcomes

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

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5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Nowadays due to increase in workload on academic studies for university students, they are constantly in a state of stress and burned out specially during examination times, because the current academic support platforms only focus on providing the content that the student must follow regardless of their mental state which can impact negatively towards a learner's efficiency and motivation. As a result, this research addresses the issues that come up when students start to study for their exams and cover up their lessons which rather than following a rigid study plan, they can follow an adaptive study plan that incorporates their emotional state to their study plan. Therefore, we are proposing a solution which incorporated students mental and emotional state when giving them a study plan to follow and adjust it accordingly. Also, this checks the correctness of the output by reducing the hallucinations of AI-generated outputs while giving students a study plan based on lesson objectives rather than on lesson content.

Currently Large Language Models (LLMs) for educational tutoring, existing AI-based study assistants focus on question answering without sufficient grounding in course-specific materials. while LLM tutors can improve conceptual understanding, but frequently suffer from hallucinated, especially when queries are outside their training distribution or lack contextual grounding [1], [5]. Retrieval-augmented generation (RAG) approaches, effective in filtering responses to relevant documents, apply static prompting strategies and not account for response appropriateness [4]. This limitation causes risk in academic ground, where incorrect explanations can mislead students and negatively impact learning outcomes. Therefore, a tutoring mechanism that not grounds responses in verified course materials but dynamically adapts explanation strategies to reduce hallucination risks and improve teaching reliability.

AI-based summarization tools and flashcard generation systems are increasingly used to help students revise large volumes of academic content. However, existing solutions generate generic summaries that prioritize textual compression over educational relevance [2], [5]. Studies shows summaries often fail to align with lecture objectives, assessment criteria, or expected learning outcomes, limiting their effectiveness as study aids [2]. Furthermore, most summarization systems do not distinguish between core examinable concepts and additional material, leading to cognitive overload. This highlights a research gap in developing learning-objective-aware summarization approaches that suit outputs for academic revision, rather than producing general-purpose summaries.

Research indicates that students' emotional and cognitive states—such as confusion, frustration, boredom, or engagement—have a significant influence on learning effectiveness, motivation, and academic performance [1], [3]. Recent multimodal tutoring systems that incorporate facial expression analysis and interaction-based affect signals demonstrate notable improvements in personalization and pedagogical effectiveness compared to emotion-agnostic systems [1]. However, many existing academic support platforms either ignore emotional factors entirely or rely on manual self-reporting, which is often inaccurate and disruptive to the learning process. Furthermore, while emotion-aware tutoring frameworks exist, they are rarely integrated into practical study-planning or exam-preparation workflows, limiting their real-world applicability [3]. This highlights the need for an automated emotion detection component that can infer students' emotional states during study sessions and provide adaptive learning support, thereby enhancing engagement and reducing cognitive overload.

Current AI-driven study planners and reminder systems mainly use static or rule-based scheduling techniques, such as spaced repetition, which adapt only to performance metrics like quiz scores or completion history [2]. While these approaches improve retention, they fail to account for learners' emotional readiness or mental fatigue, which can significantly influence adherence and motivation [3]. Research shows that cognitive overload and negative emotional states can reduce the effectiveness of reminders and increase procrastination [3]. Despite this, the integration of affective signals into adaptive reminder scheduling remains largely unexplored in existing systems [2]. This indicates a research gap in developing emotion- and progress-aware reminder mechanisms that adjust study schedules based not only on content mastery but also on students' emotional and cognitive states.

References

[1] D. Kar, L. Böss, D. Braca, S. Dennerlein, and N. Hubig,
"MathBuddy: A multimodal system for affective math tutoring," *arXiv preprint* arXiv:2508.19993, Aug. 2025.
doi: 10.48550/arxiv.2508.19993.
[2] V. Mehta and A. Sharma,
"PrepPal: Enhancing learning mechanism using AI integration," in *Proc. IEEE Int. Conf. Emerging Computing and Communication Technologies* (INECCE), 2025, pp. 622–627.
doi: 10.1109/inecce64959.2025.11150997.
[3] B. Fetaji, M. Fetaji, G. Armenski, and F. Fetaji,
"Assessing personalized engineering learning experience with a multi-modal AI tutoring framework," in *Proc. IEEE MIPRO*, June 2025, pp. 1437–1442.
doi: 10.1109/mipro65660.2025.11131798.
[4] Z. Jiang and M. Jiang,
"Beyond answers: Large language model-powered tutoring system in physics education for deep learning and precise understanding," *arXiv preprint* arXiv:2406.10934, June 2024.
doi: 10.48550/arxiv.2406.10934.
[5] Z. Liu, P. Agrawal, S. Singhal, V. Madaan, M. Kumar, and P. K. Verma,
"LPITutor: An LLM-based personalized intelligent tutoring system using RAG and prompt engineering," *PeerJ Computer Science*, vol. 11, Aug. 2025.
doi: 10.7717/peerj-cs.2991.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

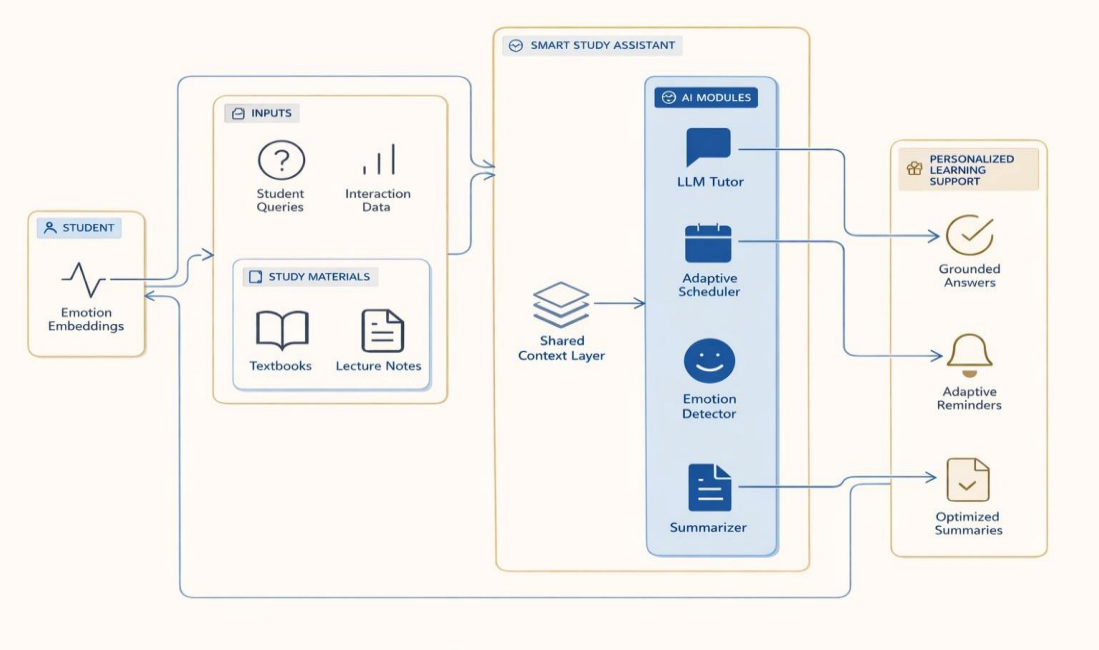
The proposed solution is an integrated Smart Study Assistant designed to provide personalized, relating to teaching grounded, and privacy-aware learning support for university students by combining multiple AI-driven components into a unified system. The system addresses limitations and weaknesses in existing study tools by including context-wise understanding, emotional awareness, and adaptive decision-making.

At the core of the system is an LLM-based tutoring module which enhanced with a confidence-aware retrieval-augmented generation (RAG) mechanism. This component answers student queries using verified course-specific materials while dynamically adjusting response strategies (direct answers, hints, or clarification requests) based on model confidence, thereby reducing hallucinations and improving trust.

A learning-objective-aware summarization module processes lecture slides and transcripts to generate hierarchical or ranked summaries and exam-oriented flashcards aligning with course outcomes. This ensures that generated content is optimized for learning effectiveness rather than generic text compression.

To enhance personalization, a privacy-preserving emotion detection module that operates on-device to extract low-dimensional affect embeddings from webcam input, discarding raw images immediately. These embeddings are used solely to capture learners' emotional states without compromising privacy.

Finally, an emotion- and progress-aware adaptive reminder system that integrates learning performance data, memory decay models, and emotional state signals to intelligently schedule study reminders that maximize adherence while minimizing cognitive fatigue.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The proposed Smart Study Assistant requires specialized expertise across multiple domains of Information Technology, with each system component demanding distinct knowledge, tools, and data sources.

LLM-Based Q&A Tutor:

This component requires expertise in Natural Language Processing, Large Language Models, and retrieval-augmented generation techniques. Knowledge of prompt engineering, models for answer evaluation is essential to generate reliable academic responses. Development tools include LLM APIs or open-source models and vector databases. The primary data required consists of course-specific academic materials such as lecture slides, textbooks, syllabus, and learning outcomes. These structured and unstructured text-based documents are indexed to ground responses in verified course content.

Lecture Notes Summarization Module:

This component requires knowledge in NLP-based summarization, instructional design, and educational content modeling. Instruction-tuned language models and document processing tools are used to generate concise summaries and revision-focused flashcards. The data required includes lecture notes, presentation slides, transcripts, and assessment guidelines. Small-scale human-evaluated summaries or flashcards may be collected from students or instructors for validation purposes.

Emotion Detection Module:

Developing this component requires expertise in computer vision and facial expression analysis. Machine learning models are used to classify emotional states during study sessions. The data required includes webcam video frames captured during voluntary and consented study sessions, along with self-reported emotional labels for basic calibration and evaluation. This data is used to improve emotion classification accuracy and adaptive feedback mechanisms.

Adaptive Reminder and Study Planning Module:

This component requires expertise in machine learning, learning analytics, and cognitive learning theories such as spaced repetition. Scheduling algorithms and adaptive decision models are employed to personalize study reminders. The data required includes student study activity logs, quiz performance records, revision history, and detected emotional states. This time-series and behavioral data enable the system to adjust reminder timing and study plans dynamically.

With all, these domain expertise areas and data sources are needed for the development of an effective, adaptive, and academically grounded Smart Study Assistant.

8. Objectives and Novelty

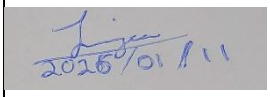
Main Objective Developing an integrated, intelligent, privacy-aware Smart Study Assistant that improves the university students' learning effectiveness by combining grounded AI tutoring, learning-objective-aware summarization, privacy-preserving emotion detection, and emotion-aware adaptive study reminders.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
I.H.S. Dilmith IT22120120	To design and implement hallucination-controlled LLM-based tutoring system grounded in course-specific materials	• Design confidence-aware RAG pipeline • Integrate course materials into vector database • Implement uncertainty-based response strategies (answer, hint, clarification) • Evaluate hallucination rate and answer accuracy	Introduces confidence-aware relating to teaching control in LLM tutoring, reducing hallucinations by adapting responses based using model certainty rather than relying on prompt heuristics
T.G.K.G. Vijithapala IT22187178	To develop learning objective aware lecture summarization and flashcard generation system	• Collect lecture slides and transcripts • Align summaries with learning outcomes • Generate rank-based summaries and exam-oriented flashcards • Evaluate usefulness through quizzes and student feedbacks.	Including relative to teaching priors and learning objectives into AI summarization, shifting focus from general summaries to assessment-relevant learning support.

K.K.G.Y. Mihiraj IT22224552	To implement an emotion detection module for affect-aware learning support	• Design emotion detection pipeline • Extract low-dimensional affect embeddings • Discard raw visual data immediately • Evaluate emotion accuracy and perception	Proposing an affective computing approach, using emotion embeddings instead of storing raw facial data which is an ethical advancement in emotion-aware tutoring.
W.H.H. Thathsara IT22371690	To design an emotion and progress aware adaptive study reminder system	• Model study progress and memory decay • Integrate emotional state signals • Implement adaptive reminder scheduling logic • Evaluate adherence(attention), retention, and fatigue	First integration of emotional readiness into adaptive study scheduling, extending traditional spaced repetition with affect-aware decision making.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
I.H.S. Dilmith IT22120120	This component aligns with the specialization in **Artificial Intelligence and Software Engineering**, focusing on Natural Language Processing and Large Language Models. The member applies domain knowledge in LLMs, retrieval-augmented generation (RAG), embedding models, and uncertainty estimation to develop a grounded and hallucination-controlled AI tutoring system. The work demonstrates applied AI specialization through model integration, system evaluation, and reliability enhancement in educational contexts.
T.G.K.G. Vijithapala IT22187178	This component complies with the specialization in **Data Science and Intelligent Systems**, particularly NLP-based text processing and educational analytics. The member leverages expertise in text summarization, information extraction, and instructional design to align AI-generated summaries with learning objectives. The component reflects specialization by combining pedagogical knowledge with advanced NLP techniques to optimize learning effectiveness rather than generic summarization.
K.K.G.Y. Mihiraj IT22224552	This component aligns with the specialization in **AI, Computer Vision, and Ethical Computing**. The member applies skills in image processing, affective computing, and machine learning to design an on-device emotion detection system. By avoiding raw data storage and using low-dimensional embeddings, the component demonstrates responsible AI development consistent with ethical and privacy-focused computing specialization.
W.H.H. Thathsara IT22371690	This component complies with the specialization in **Intelligent Systems and Applied Machine Learning**, emphasizing decision-making and adaptive systems. The member applies knowledge of spaced repetition, predictive modeling, and user behavior analysis to design an emotion- and progress-aware adaptive reminder system. The integration of emotional signals into scheduling logic demonstrates advanced specialization in human-centered intelligent system design.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Suriya	Kumari	
Co-Supervisor	Mr.	Ravi	Supunya	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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Topic Assessment Form

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.